

Grading Checklist

Course: CMPS323 Design and Analysis of Algorithms

Assignment: Multi-Way Merge Sort

Total: 100 points

Student Name: _____ ID: _____

1. Basic Requirements (Mandatory)

- ☐ Uses divide and conquer and recursion
- ☐ No built-in sorting functions used
- ☐ Code compiles and runs

Major violations may result in zero for the affected section.

2. Three-Way Merge Sort (30 pts)

- ☐ Correct divide step (three subarrays) – 6 pts
- ☐ Correct recursive structure – 6 pts
- ☐ Correct merge logic – 6 pts
- ☐ Handles edge cases (uneven splits, duplicates) – 6 pts
- ☐ Produces correct output – 6 pts

Score: _____ / 30

3. Four-Way Merge Sort (30 pts)

- ☐ Correct divide step (four subarrays) – 6 pts
- ☐ Correct recursive structure – 6 pts
- ☐ Correct merge logic – 6 pts
- ☐ Handles edge cases (subarray exhaustion) – 6 pts
- ☐ Produces correct output – 6 pts

Score: _____ / 30

4. Experimental Setup (15 pts)

- ☐ Warmup runs implemented – 3 pts
- ☐ Warmups in required range (5–10) – 3 pts
- ☐ Timed runs in required range (5–10) – 3 pts
- ☐ Input sizes cover required range (50k to 1.6M) – 3 pts
- ☐ Intermediate sizes chosen independently – 3 pts

Score: _____ / 15

5. Analysis and Discussion (25 pts)

- ☐ Time complexity derived for 3-way merge sort – 5 pts
- ☐ Time complexity derived for 4-way merge sort – 5 pts
- ☐ Experimental results clearly presented – 5 pts
- ☐ Performance trends clearly described – 5 pts
- ☐ Explanation relates observations to algorithm design – 5 pts

Score: _____ / 25

6. Code Quality (Penalty-Based)

- ☐ Clear structure and readable code
- ☐ Reasonable variable and method names
- ☐ Comments explain non-trivial logic

Up to **-5 points** may be deducted for very poor readability.

Penalty: _____

7. Similarity Check (TA Use Only)

- ☐ Identical merge logic structure
- ☐ Identical experimental parameters
- ☐ Identical comments or variable names

If suspicious, flag for review.

Final Score

Subtotal: _____ / 100

Penalty: _____

Final Grade: _____ / 100

TA Name: _____ Date: _____